

Smart File Organizer

Python Programming Task 08

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Internship: Python Programming Internship

1. Introduction

The Smart File Organizer is a Python application that automatically organizes files into separate folders based on their file extensions. It helps users manage files efficiently by reducing clutter and improving accessibility. The project demonstrates Python automation, file handling, directory management, and exception handling.

2. Objective

Develop an automated file management system that scans a selected directory, categorizes files, organizes them into appropriate folders, generates statistics, detects duplicate file names, and creates a report.

3. Tools & Technologies Used

- Python3
- os Module
- shutil Module
- datetime Module
- collections Module

4. Modules Implemented

Module 1 – Directory Selection

Accept folder path from the user, validate the directory, and handle invalid paths.

Module 2 – File Scanning

Scan all files, display file names, extensions, and total files.

Module 3 – Automatic File Organization

Organize files into Images, Documents, Videos, Audio, Archives, Programs, and Others.

Module 4 – File Statistics

Display category-wise file counts.

Module 5 – Search Functionality

Search by file name and extension.

Module 6 – Duplicate File Detection

Detect duplicate file names and display results.

Module 7 – Report Generation

Generate file_report.txt containing date, folder name, statistics, duplicates, and folder structure.

Module 8 – Exception Handling

Handle invalid folder, missing folder, permission denied, file already exists, and unexpected errors.

5. Project Workflow

1. User enters folderpath.
2. Directory is validated.
3. Files are scanned.
4. Files are categorized by extension.
5. Required folders are created.
6. Files are moved.
7. Statistics are displayed.
8. User searches files.
9. Duplicate files are checked.
10. Report is generated.

6. Output

The application automatically organizes files, creates categorized folders, displays statistics, searches files, detects duplicate file names, and generates a detailed report.

7. Challenges Faced

Managing different file extensions, moving files safely, handling invalid directories, and implementing exception handling.

8. Conclusion

The Smart File Organizer automates file management by organizing files into categorized folders, generating statistics, detecting duplicate file names, and producing a detailed report. It demonstrates practical Python skills in automation, file handling, and directory management.